

AAROH (■■■■■)

Hybrid Classical-Quantum Clinical Benchmarking & Intelligence Platform

Domain: Healthcare AI / Oncology	Dataset: WBCD (569 Biopsies, 30 Features)	Stack: Qiskit Aer, XGBoost, TreeSHAP, FastAPI, React 19
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EXECUTIVE SUMMARY (THE 60-SECOND TL;DR)

What is AAROH in simple words?

AAROH is a medical decision-support platform for breast cancer biopsy diagnosis. It runs **BOTH** a production classical model (XGBoost) and a 4-qubit Quantum Classifier (Qiskit VQC) through an auditable, real-time agent pipeline. It explains every decision with exact TreeSHAP attribution forces and generates an official printable hospital pathology memo.

1. The Core Pitch & The 'Honest Numbers' Strategy

Many hackathon teams lose credibility by making fake '99.9% Quantum Advantage' claims. AAROH takes the winning scientific path:

THE OFFICIAL ELEVATOR PITCH (MEMORIZE FOR FACULTY & JUDGES)

'AAROH is a hybrid clinical benchmarking platform that orchestrates classical XGBoost and a trained Variational Quantum Classifier through an agentic pipeline, producing SHAP-explainable predictions and a clinical consultation memo — with the honest result that at 569 rows, we validate quantum encoding viability, not quantum superiority.'

- Three Foundational Pillars to Remember:
- **Real Computation Under the Hood:** Zero hardcoded metric arrays, zero mocked latency timers, zero hallucinated SHAP values. Every tensor is computed live.

• **The Honest Numbers Protocol:** On 569 tabular rows, classical XGBoost is mathematically optimal (0.994 ROC-AUC). Our 4-qubit VQC achieves 0.748 ROC-AUC on a held-out test split, validating non-linear Hilbert space encoding viability under NISQ quantum constraints.

• **Deterministic Agentic Telemetry:** Instead of unpredictable LLMs that can hallucinate medical dosages, our 4-stage pipeline is 100% deterministic, auditable, and streamed live via Server-Sent Events (SSE).

2. System Architecture & Component Roles

Subsystem	Tech Stack	Role & Key Responsibilities
Backend API (Port 8000)	Python 3.12, FastAPI, Uvicorn	Serves REST & SSE endpoints. Manages lifespan model pre-warming so inferences respond in sub-70 milliseconds.
Hybrid ML & Quantum Core	Qiskit Aer, XGBoost, TreeSHAP, NumPy	Executes 5-Fold CV XGBoost, 4-qubit ZZFeatureMap + RealAmplitudes VQC, zero-leakage TreeSHAP, and zero-fail NumPy fallback matrix simulator.
Deterministic Orchestrator	Python Generators, Server-Sent Events	Streams stage execution telemetry in real-time packets (Stage 1 → Stage 2 → Stage 3 → Stage 4 → Final Result).
Clinical Frontend UI (Port 5173)	React 19, Vite, Tailwind CSS, Lucide, Recharts	Glassmorphic clinical dark theme, WBCD presets, live agent stream terminal, interactive SHAP force bars, and 1-click printable hospital memo.

3. Step-by-Step Diagnostic Pipeline (The 4 Agent Stages)

When a user clicks 'Run Hybrid Diagnostic Pipeline', here is the exact 4-stage sequence streamed live via SSE:

Stage	Agent Name	Exact Computation Performed
Stage 1	Data Ingestion & QC Agent	Validates patient vector has exactly 30 positive measurements. Checks bounds, zero-NaN verification, and logs nuclear morphology baseline (Radius, Perimeter, Area).

Stage 2	Quantum Encoding Agent	Compresses 30 features to 4 PCA components, maps to $[-\pi, +\pi]$ angles, encodes into 4 qubits via ZZFeatureMap (reps=1), applies RealAmplitudes ansatz (8 parameters), and measures $\langle IIIZ \rangle$ to calculate quantum probability.
Stage 3	Classical Explainability Agent	Executes XGBoost inference for calibrated confidence and log-odds margin $f(x)$. Runs exact TreeExplainer feature attributions, verifying additive invariant: $f(x) = E[f(x)] + \sum \phi_i$ with zero margin leakage ($\delta < 1e-6$).
Stage 4	Clinical Synthesis & Memo Agent	Cross-checks dual-model concordance. Assigns ICD-10 coding (C50.919 malignant carcinoma or N60.99 benign dysplasia). Assigns risk tier and synthesizes a printable hospital consultation memo with clinical recommendations.

4. Benchmark Metrics & Faculty / Judge Defense Sheet

Memorize these exact empirical numbers and defense scripts for presentations:

Metric	Classical XGBoost (5-Fold CV)	4-Qubit VQC (80/20 Test Split)	Clinical Interpretation
ROC-AUC	0.994 ± 0.004	0.748	Separation power between malignant and benign cases.
Accuracy	96.0% ± 1.6%	68.4%	Overall correct classification rate across test samples.
Sensitivity (Recall)	93.4% ± 3.3%	69.1%	Critical in oncology: Percentage of actual cancers caught.
Specificity	97.5% ± 1.3%	68.1%	Percentage of benign cases correctly ruled out.
Sample Size	n = 569 (All cases)	n = 114 (Held-out test split)	Evaluation asymmetry disclosed via Honest Protocol.

THE 3 PRE-WRITTEN FACULTY / JUDGE RESPONSES

Q1: 'Why use Quantum ML when XGBoost already gets 99% ROC-AUC?'

Answer: 'At 569 tabular rows, classical ML is mathematically optimal. AAROH does not claim quantum supremacy today; it is a benchmarking platform designed to validate non-linear quantum Hilbert space feature encoding viability and provide clinical audibility as quantum processors scale toward fault tolerance.'

Q2: 'How are quantum states represented and measured?'

Answer: 'Features are mapped to 4 qubits via a second-order Pauli expansion (ZZFeatureMap), parameterized by an entangling RealAmplitudes ansatz (8 parameters), and measured through the Z-basis expectation value $\langle IIIZ \rangle$ on qubit 0.'

Q3: 'What if Qiskit fails or simulator crashes during live evaluation?'

Answer: 'AAROH implements a zero-fail multi-tier fallback: Level 0 runs live Qiskit Aer; Level 1 automatically routes to our pre-computed pure NumPy statevector matrix simulator (0.000000 delta parity in <4ms); Level 2 defaults to full classical XGBoost with honest diagnostic disclaimer.'

5. Explainable AI (XAI) & The Doctor's Trust Layer

Doctors reject black-box AI. AAROH makes every prediction 100% transparent using exact **TreeSHAP**:

- **Base Value $E[f(x)] = -0.5699$:** The baseline probability of cancer across the general cohort.
- **Attribution Forces:** Each of the 30 cell nucleus measurements pushes the score either toward cancer (Positive / Red force) or toward benign (Negative / Green force).
- **#1 Clinical Cancer Driver:** *Worst Concave Points* (severe indentation of cell nucleus margins, a hallmark sign of carcinoma).
- **Formal Pathology Memorandum:** Generates ICD-10 diagnostic codes (C50.919 / N60.99) and recommended next clinical steps (e.g. Core Needle Biopsy with IHC profiling ER/PR/HER2). Clicking 'Print' renders a clean black-and-white hospital sheet ready for clinical records.

6. Codebase Tour — Where Everything Lives

REPOSITORY MAP

backend/main.py: FastAPI endpoint, CORS, lifespan model cache, /api/predict/stream.

backend/agents/orchestrator.py: 4-stage deterministic SSE generator & input validator.

backend/models/classical.py: XGBoost trainer with 5-Fold Stratified CV.

backend/models/quantum.py: 4-Qubit VQC circuit (ZZFeatureMap + RealAmplitudes).

backend/models/fallback_sim.py: Pure NumPy matrix simulator with zero dependencies.

backend/models/explainability.py: TreeSHAP feature attributions & narrative builder.

backend/reports/memo_gen.py: ICD-10 pathology memorandum synthesizer.

aaroh-app/src/App.jsx: Master React dashboard handling live SSE streams.

aaroh-app/src/components/: Modular UI components (PatientInput, AgentStream, ModelComparison, ShapViewer, ClinicalMemo, MetricsModal).

idea.md & progress.md: Full pitch scripts, Q&A; defenses, and master engineering logs.

7. How to Run AAROH Locally (Quickstart for the Team)

LOCAL SETUP COMMANDS

Terminal 1 — Backend Service (Port 8000):

cd SIH_Project2026
backend\.venv\Scripts\activate (or source backend/.venv/bin/activate)
pip install -r backend/requirements.txt
python -m uvicorn backend.main:app --host 127.0.0.1 --port 8000 --reload

Terminal 2 — Frontend UI (Port 5173):

cd aaroh-app
npm install
npm run dev

Open browser at **http://localhost:5173/**. Select 'Malignant' or 'Benign' preset and click 'Run Hybrid Diagnostic Pipeline'.

8. Future Roadmap: How We Can Elevate This Further

High-impact directions the team can take to make AAROH even more impressive:

Enhancement Area	Specific Technical Task	Expected Impact on Judges
1. Frontend 'De-AI-fying' & Clinical Polish	- Refine UI to look like an authentic hospital pathology workstation (Cerner/Epic EHR style). - Add interactive SVG cell cytology diagrams showing cell margin irregularity. - Add dark/light mode hospital toggle.	Removes any 'generic template' feel. Proves deep domain empathy for how pathologists actually review tissue biopsies.
2. Multi-Modal Histology Imaging	- Integrate Whole Slide Images (WSI) alongside numerical FNA tables. - Train a lightweight Vision Transformer (ViT) or ResNet feature extractor and concatenate image embeddings with 30D morphometry.	Elevates project from purely tabular data into cutting-edge multi-modal AI, a top criteria in SIH healthcare tracks.
3. Scaling to Real IBM Quantum Hardware (QPU)	- Connect backend to IBM Quantum Runtime API (via free IBM Cloud API token). - Execute circuits on real superconducting 127-qubit processors (Eagle/Heron) with readout error mitigation (M3/TREX).	Demonstrates real physical quantum execution rather than just simulation, validating true NISQ capability.
4. Privacy-Preserving Federated Learning	- Simulate 3 distinct hospital nodes (e.g. AIIMS, Tata Memorial, Apollo). - Train models using federated averaging (FedAvg) with differential privacy so patient data never leaves hospital servers.	Addresses major real-world regulatory hurdles (HIPAA / GDPR / DISHA compliance) for hospital AI adoption.

5. Multi-Cancer Generalization	Adapt pipeline to other cytology datasets: Thyroid FNA (Bethesda), Lung adenocarcinoma, or Melanoma dermoscopy.	Proves AAROH is a universal clinical orchestration engine, not just a one-dataset demo.
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9. Presentation Day Cheat Sheet (Key Terms Decoded)

- FNA (Fine Needle Aspiration):** A quick biopsy technique where a thin needle extracts cell clusters from a breast lump.
- Ansatz:** A parameterized quantum circuit with adjustable rotation gates (like weights in a neural network).
- ZZFeatureMap:** A quantum mapping technique that entangles pairs of qubits to encode non-linear feature correlations.
- Observable <IIIZ>:** Measuring the Pauli-Z operator on qubit 0 to get an expectation value between -1.0 and +1.0.
- TreeSHAP:** An exact algorithm by Scott Lundberg that computes the mathematical contribution of each feature to the tree margin.
- ICD-10 C50.919:** Official International Classification of Diseases code for malignant breast neoplasm.

Repository Link: https://github.com/CodeCrafterAdi2006/AAROH_Platform123
Prepared for: AAROH Engineering Team & Evaluators | Smart India Hackathon (SIH 2026)